DOC-18 | CONTROLLED SOURCE SYNCHRONIZATION

# Cross-reference-only revision notice

**v1.0-RC3** | 11 September 2026 | SOURCE-OF-TRUTH SYNCHRONIZATION ONLY

| Control | Selected value |
|---|---|
| **Revision** | v1.0-RC2 -> **v1.0-RC3** |
| **Semantic change** | **NONE.** This revision changes source version/count/path references only. |
| **Wave-2 semantic authorities** | DOC-07 RC2 \| DOC-08 RC2 \| DOC-15 RC8 \| DOC-16 RC5 |
| **Canonical HTTP companion** | contracts/http/BLOG-DOC-15_openapi_v1.0-RC8.yaml |
| **OpenAPI SHA-256** | bf577a463b4c5d8fdceddc2087f53216426ef88d67d8e1a0d22ccf819751cd00 |
| **RC8 estate** | 113 operations \| 86 paths \| 221 schemas/helpers \| 1,311 resolved internal refs \| 0 unresolved refs<br>12 AUTH \| 2 PUBLIC/PREVIEW \| 63 SITE \| 36 GLOBAL |

## DOCUMENT-SPECIFIC SYNCHRONIZATION

• Cross-reference update only: HTTP authority is DOC-15 v1.0-RC8 and SEO/search/sitemap authority is DOC-16 v1.0-RC5.

• Worker execution estate remains exactly 6 logical queues, 11 canonical job types and 7 retry profiles. No job, queue, Redis key semantic or worker behavior changes.

## SUPERSESSION AND HISTORICAL CLASSIFICATION

The inherited body that follows is retained unchanged in substantive page content. Any inherited non-owning reference to DOC-15 RC7/RC6, the old 112/85/217/1,297 estate, 11 AUTH operations, DOC-16 RC4, or the RC7 YAML filename is classified **HISTORICAL / SUPERSEDED REFERENCE** and is not normative in this revision. All other inherited owned semantics remain in force.

Occurrence ledger: RC4: inherited printed pages 1, 5, 27; pre-RC8 HTTP upstream label: inherited printed pages 1.

**IMPLEMENTATION IMPACT: NONE. The accepted NAKSHATRA_MASTER_WAVE2_HANDOFF application/database/API behavior is unchanged by this document synchronization.**

BLOG-DOC-18

# Workers, Scheduling, Queue, Redis Cache & Background Processing Specification

Deterministic asynchronous execution, delayed publishing, derived-system convergence and tenant-safe Redis/cache mechanics for the centralized 14-site blogging platform

| 6 | 11 | 7 | 14 |
|---|---|---|---|
| logical queues | canonical job types | retry profiles | site identities |

**CORE BACKGROUND-PROCESSING PRINCIPLE**

PostgreSQL/content state remains authoritative. Queues, Redis caches, search documents, sitemap artifacts and media derivatives are rebuildable/derived execution state. Every site-owned job carries one trusted site identity, revalidates current ownership/lifecycle/state before side effects, and is safe under at-least-once delivery.

| Document control | Value |
|---|---|
| Status | REVIEW CANDIDATE - not frozen until approved |
| Version | 1.0-RC2 |
| Date | 10 September 2026 |
| Frozen upstream authorities | BLOG-DOC-01 through BLOG-DOC-14; BLOG-DOC-15 v1.0-RC6; BLOG-DOC-16 v1.0-RC4; BLOG-DOC-17 v1.0-RC2 |
| Primary authority | Logical queue registry; job envelope/payload contracts; scheduling/reconciliation; retry/backoff/timeouts; idempotency/deduplication; concurrency/leases; dead-letter/replay; Redis namespace/key/TTL/invalidation conventions; background-processing observability. |
| Explicit non-authority | Product features/permissions (01-03), tenant model (04), content lifecycle (07), DB DDL (08), routing (09), page flows (10-11), public data (13), frontend (14), HTTP transport (15), SEO/search/sitemap computation (16), media transform/CDN semantics (17), security implementation (19), performance/DR budgets (20), QA master suite (21), repository/runtime stack (23), deployment (24). |

## Developer lock target

A competent developer must not have to guess which queue receives a background effect, what tenant/entity identity a job carries, when a delayed schedule becomes stale, which failures retry, the exact backoff/timeout profile, how duplicate jobs collapse, how Redis keys prevent tenant collisions, how full reindex and incremental indexing serialize, how sitemap builds coalesce, or how dead-letter replay remains site-safe.

# 0. Document Map


| Section | Subject |
|---|---|
| 1 | Purpose, authority and conflict rules |
| 2 | Frozen upstream baseline and inherited invariants |
| 3 | Logical background-processing architecture |
| 4 | Execution contexts, trust and authority |
| 5 | Common job envelope and identity |
| 6 | Canonical queue registry and delivery semantics |
| 7 | Producer ordering, durable truth and enqueue recovery |
| 8 | Scheduling architecture and delayed-work recovery |
| 9 | Scheduled publishing job contract |
| 10 | Media-processing job contract |
| 11 | Incremental public-search indexing |
| 12 | Full site reindex contract |
| 13 | XML sitemap generation contract |
| 14 | Cache invalidation worker contract |
| 15 | Maintenance/reconciliation jobs |
| 16 | Failure classification and retryability |
| 17 | Retry, exponential backoff and attempt timeouts |
| 18 | Idempotency and deduplication |
| 19 | Concurrency, serialization, leases and stale workers |
| 20 | Dead-letter queues and replay |
| 21 | Redis namespace and exact key grammar |
| 22 | Redis cache classes, values and TTLs |
| 23 | Dependency tracking and invalidation rules |
| 24 | Search-status coordination in Redis |
| 25 | Sitemap dirty/build coordination |
| 26 | Queue ordering, priority and fairness |
| 27 | Explicit global fan-out and site enumeration |
| 28 | Site lifecycle/domain-change background effects |
| 29 | Failure/recovery state matrix |
| 30 | Logging, metrics and correlation |

| Section | Subject |
|---|---|
| 31 | Security/privacy boundaries |
| 32 | End-to-end execution examples |
| 33 | Cross-document consistency audit |
| 34 | Traceability registry |
| 35 | Comprehensive acceptance criteria |
| 36 | Dependencies, downstream handoff and freeze-readiness |
| A | Canonical job registry |
| B | Queue/job/retry matrix |
| C | Redis key registry |
| D | Trigger-to-job matrix |
| E | Failure/retry quick matrix |
| F | One-page developer execution reference |

**AUTHORITY BOUNDARY**

DOC-18 owns asynchronous execution mechanics and Redis/cache mechanics only. It consumes business truth from earlier documents. It may define a job, queue or cache key needed to implement an already-approved effect, but it may not create a new user-visible capability, permission, endpoint, page, database table/column, content state, media variant, search rule or sitemap/redirect algorithm.

# 1. Purpose, Authority and Conflict Rules

## 1.1 Purpose

BLOG-DOC-18 converts the already-frozen asynchronous side effects from publishing, media, search, sitemap and cache authorities into one deterministic worker/queue/cache execution contract. The document makes those effects retry-safe and tenant-safe without moving authoritative business state out of PostgreSQL or redefining upstream semantics.

## 1.2 One source of truth per decision

| Decision | Authority |
|---|---|
| What features/capabilities exist | DOC-03 |
| Who may request protected operations | DOC-02 |
| Trusted site/global/worker context | DOC-04 |
| Post schedule/publish lifecycle | DOC-07 |
| Authoritative persistence/version/RLS | DOC-08 |
| Admin/public flow effects | DOC-10 / DOC-11 |

| Decision | Authority |
|---|---|
| HTTP status/accepted-command contract | DOC-15 |
| SEO/search/sitemap/redirect computation | DOC-16 |
| Media processing/variant/CDN semantics | DOC-17 |
| Queue/job/retry/Redis/cache execution | DOC-18 - this document |

### 1.3 Conflict rule

**CONFLICT RULE**

If a worker design would require a new permission, API endpoint, persistent jobs/outbox table, content lifecycle state, media variant key, search ranking rule, sitemap resource, cross-site ownership model or other upstream semantic change, implementation stops. DOC-18 must either use existing authority or record a genuine upstream gap; it cannot silently solve the conflict.

## 2. Frozen Upstream Baseline and Inherited Invariants

| Upstream | DOC-18 inherited requirement |
|---|---|
| DOC-01 | One logical worker system may be shared; site-owned background work and cache/search/sitemap artifacts remain isolated. Exact cache keys/job payloads are delegated here. |
| DOC-02 | System-derived work does not create permissions. User-triggered reindex is authorized before acceptance; workers do not treat a queued payload as a permission token. |
| DOC-03 | Scheduled publishing, search/reindex, media system variants and related system effects are approved capabilities; no generic automation feature is added. |
| DOC-04 | CTX-WORKER-SITE; minimum semantic job identity is site_id + target + job/action + idempotency identity; stale jobs revalidate site/target/state; retries/DLQ preserve tenant. |
| DOC-07 | SCHEDULED is authoritative in PostgreSQL; reschedule creates a new version/generation; scheduled worker revalidates due time/status/version and then performs the frozen publish transition. |
| DOC-08 | No queue/cache/jobs table exists; posts.version and schedule fields are authoritative; RLS/site_id remain enforced in worker transactions. |
| DOC-09 | Canonical host/route/redirect rules are not recomputed here; cache/sitemap workers consume current route authority. |
| DOC-10/11 | Reindex is an accepted site-aware system effect; public stale/wrong-site data is never substituted during failures. |
| DOC-13 | PageData never exposes queue/cache/job IDs/keys; derived caches must project canonical data rather than raw DB rows. |

| Upstream | DOC-18 inherited requirement |
|---|---|
| DOC-14 | Fourteen independent frontends do not share mutable runtime tenant state; frontend applications do not consume internal Redis/job contracts. |
| DOC-15 RC6 | API15-S-094 is 202 AcceptedCommand with no job ID; Admin/Preview success is no-store; API idempotency is 24h and separate from worker dedupe. |
| DOC-16 RC4 | Public search index is rebuildable same-site derived state; reindex is site-scoped; sitemap resource/eligibility/sharding/order and search ranking are fixed upstream. |
| DOC-17 RC2 | Native WebP baseline is sanitized/public; remaining variants may complete asynchronously; exact job payload/retry and cache invalidation are delegated here. |

**DERIVED-STATE DOCTRINE**

Redis, queue records, search index documents, sitemap generations and media variant work are never the canonical source for editorial truth. Loss of derived state must degrade service or trigger reconstruction; it must not rewrite PostgreSQL truth from stale worker payloads.

# 3. Logical Background-Processing Architecture

| Authoritative mutation / command | Worker layer | Derived target |
|---|---|---|
| Schedule/Reschedule commit | Delayed scheduler + publish queue | Future DOC-07 publish transition |
| Upload/Replace/current media generation | Media queue | Remaining DOC-17 fixed variants + cache/CDN effects |
| Public-affecting content/taxonomy/author changes | Search queue | DOC-16 public search index |
| SEO/public eligibility/domain changes | Sitemap queue | DOC-16 site-local sitemap generation |
| Public data/media/site-lifecycle changes | Cache queue | Tenant-keyed Redis + required CDN purge adapter |
| Periodic convergence checks | Maintenance queue | Repairs missing derived work from authoritative state |

Logical topology:

```
PostgreSQL / authorized API commit
        |
        +--> queue producer (site-aware) --> logical queue --> CTX-WORKER-SITE
                                                  |
                                                  +--> re-read authoritative state under RLS
                                                  +--> idempotent derived side effect
                                                  +--> Redis status/invalidation/dedupe
                                                  +--> site-aware logs/metrics

Periodic reconciliation reads authoritative state and repairs any enqueue/derived-state gap.
```

- Queue provider, programming language, worker library and concrete process topology are later engineering choices under DOC-23; the logical queue names and semantics in DOC-18 are invariant.
- Workers may scale horizontally, but concurrency keys in this document must still prevent two workers from committing conflicting side effects for the same tenant/entity/generation.
- No frontend may connect directly to Redis or a queue broker.

# 4. Execution Contexts, Trust and Authority

| Context | Allowed use | Required revalidation |
|---|---|---|
| CTX-WORKER-SITE | Any job reading/mutating site-owned data or derived site state. | Resolve site_id; verify lifecycle; SET LOCAL tenant context; reload entity; verify entity.site_id; re-evaluate current state/eligibility. |
| CTX-GLOBAL | Frozen DOC-04 explicit global context used only by scheduler/maintenance coordination to read the global Site Registry and enumerate sites deliberately. | Never inferred from null site. Coordinator does not process site-owned rows under CTX-GLOBAL; it emits one site-scoped job and each job enters CTX-WORKER-SITE before tenant data access. |
| No context | None for site-owned work. | Fail closed; dead-letter security/invariant breach if payload is malformed or tenant cannot be proven. |

## 4.1 Worker authority is not payload authority

- `site_id`, entity IDs, version/generation hints and trigger reason in a payload are routing hints that must be verified against authoritative state; they are not authorization.
- For durable scheduled publication, system authority comes from the currently valid SCHEDULED state and due schedule after revalidation - not from the original editor session.
- For API15-S-094 reindex, DOC-15 performs user authorization before 202 acceptance. The worker then runs as an internal site-scoped system effect; the original Admin identity may be retained only for correlation/audit and never grants broader access.
- A retry or dead-letter replay must never resolve an object by entity ID alone and then adopt whatever site owns it.

# 5. Common Job Envelope and Identity

Every queued message uses the following transport-neutral logical envelope. A concrete queue library may encode it as JSON, MessagePack or native attributes, but the fields and semantics are fixed.

| Field | Presence | Contract |
|---|---|---|
| schema_version | REQ | Integer 1 for this contract. Unknown major versions are not executed. |
| job_type | REQ | One canonical JOB18-* type from Appendix A. |
| site_id | REQ for site jobs | Opaque UUID; must match the site reloaded from Site Registry and target ownership. |

| Field | Presence | Contract |
|---|---|---|
| entity_type | COND | POST, MEDIA, CATEGORY, TAG, AUTHOR, SITE or NONE according to job type. |
| entity_id | COND | Opaque target UUID when one entity is addressed. Never used without site_id. |
| source_version | COND | Post version or other upstream version when stale-generation detection requires it. |
| source_fingerprint | COND | Job-specific deterministic source-change fingerprint. Required for media generation and both incremental search job types; never public and never treated as authorization. |
| not_before | OPT | Absolute UTC instant for delayed execution. Queue must not execute earlier. |
| trigger | REQ | Frozen reason enum describing the already-approved upstream event; it does not change business semantics. |
| idempotency_identity | REQ | Deterministic internal identity defined by job type; not the public API idempotency key. |
| correlation_id | REQ | Propagated request/system correlation ID; maximum 128 chars; not secret. |
| causation_id | OPT | Prior correlation/event ID for chained derived effects. |
| enqueued_at | REQ | UTC timestamp assigned by producer. |
| payload | REQ | Job-type-specific object from Sections 9-15; unknown properties rejected by the worker contract. |

**NO PUBLIC EXPOSURE**

Job envelopes, queue IDs, Redis keys, source fingerprints and attempt metadata are internal. DOC-13 forbids them in public PageData, and DOC-15 AcceptedCommand deliberately exposes only accepted/request correlation semantics.

# 6. Canonical Queue Registry and Delivery Semantics

| Queue ID | Logical name | Responsibility | Delivery/concurrency |
|---|---|---|---|
| Q18-PUBLISH | q18.publish | Scheduled publish and due-publication recovery | At-least-once; delayed capable; site/post serialization |
| Q18-MEDIA | q18.media | DOC-17 derivative generation/repair | At-least-once; site/media/generation serialization |
| Q18-SEARCH | q18.search | Incremental public index sync + full site reindex | At-least-once; site/post or site-wide lock |
| Q18-SITEMAP | q18.sitemap | Site-local XML sitemap generation | At-least-once; one build per site at a time; coalesced |

| Queue ID | Logical name | Responsibility | Delivery/concurrency |
|---|---|---|---|
| Q18-CACHE | q18.cache | Redis invalidation and required media-CDN purge/revocation | At-least-once; idempotent deletion/epoch advance |
| Q18-MAINT | q18.maintenance | Periodic reconciliation of schedules/media/search/sitemap derived state | At-least-once; explicit site scope; no destructive retention policy |

### 6.1 Queue delivery rules

- At-least-once delivery is the required semantic. DOC-18 does not claim exactly-once execution.
- A message is acknowledged only after the worker has either committed its required side effect or classified the job as an intentional stale/no-op terminal outcome.
- Worker crash/lease expiry before acknowledgement makes the message eligible for redelivery.
- Queue storage for accepted work must survive ordinary process restart. API15-S-094 may return 202 only after its reindex command is durably accepted by the queue/dedupe layer.
- Delayed queue loss cannot become lost schedule truth: PostgreSQL SCHEDULED rows remain authoritative and the schedule reconciler repairs missing delayed messages.

## 7. Producer Ordering, Durable Truth and Enqueue Recovery

1. Validate/authorize the upstream action under the already-frozen API/service rules.
2. Commit authoritative PostgreSQL mutation first when the upstream action is a durable content/site state change. No worker observes uncommitted intent.
3. After commit, derive the minimal site-aware job(s) from the committed result and enqueue them with deterministic idempotency identities.
4. If enqueue fails after a committed mutation, record operational failure and rely on the reconciliation/TTL contracts in DOC-18 to converge; never roll back by inventing a compensating business state.
5. Exception: an endpoint whose semantic success is itself “background command accepted” (API15-S-094) must not return 202 until the durable queue acceptance succeeds. Existing DOC-15 503 behavior applies if the dependency is unavailable before acceptance.
6. Exception: DOC-17 native_webp baseline work required before media upload/replace success remains synchronous to that upstream success boundary; DOC-18 queues the remaining/repair work only.
7. A Redis dedupe CLAIMED record is only a short enqueue claim, never proof that a queue message exists. Section 18.3 defines how durable broker acceptance promotes it to ENQUEUED and how failed/crashed enqueue attempts become reclaimable.

**NO OUTBOX TABLE**

DOC-08 intentionally contains no queue/cache/jobs/outbox table. DOC-18 therefore achieves convergence through durable source-state revalidation, at-least-once queues, deterministic dedupe, short cache TTLs and periodic site-scoped reconciliation - without adding persistence structures.

## 8. Scheduling Architecture and Delayed-Work Recovery

### 8.1 Authoritative schedule identity

A scheduled publication is not identified by an extra jobs-table row. The authoritative schedule generation is the existing Post tuple:

```
schedule_generation = (site_id, post_id, posts.version, scheduled_for_utc)

Reschedule increments posts.version and changes the schedule tuple.
An older delayed job therefore becomes stale without needing a new DB column.
```

## 8.2 Producer + reconciliation model

| Mechanism | Rule |
|---|---|
| Immediate delayed enqueue | After Schedule/Reschedule commit, enqueue JOB18-PUBLISH-DUE with not_before=scheduled_for_utc and the current tuple. |
| Scheduler coordinator | Explicit CTX-GLOBAL coordinator enumerates eligible sites; it never processes site-owned posts without switching into one CTX-WORKER-SITE. |
| Site schedule reconcile | Every 60 seconds, enqueue/execute JOB18-SCHEDULE-RECONCILE-SITE for each ACTIVE site. It finds due SCHEDULED posts and ensures a deduped due job exists. |
| Clock source | Compare absolute UTC instants; editor timezone is traceability input only. Worker never re-resolves local time. |
| Early/unschedule/archive protection | Existing delayed messages are allowed to remain physically queued; state/version revalidation turns them into safe stale no-ops. |

# 9. Scheduled Publishing Job Contract

| Contract item | Locked rule |
|---|---|
| Job type | JOB18-PUBLISH-DUE |
| Queue | q18.publish |
| Required payload | site_id, post_id, source_version, scheduled_for_utc, correlation_id, idempotency_identity |
| Idempotency identity | publish:{site_id}:{post_id}:v{source_version}:{scheduled_for_utc_epoch_ms} |
| Serialization key | site_id + post_id |
| Attempt timeout | 60 seconds |
| Retry profile | R18-PUBLISH |
| Success effect | Exact DOC-07 WF-008 scheduled publish transition; then emit site-scoped cache/search/sitemap derived work. |
| Terminal stale/no-op | Site not ACTIVE; Post missing/wrong-site; status no longer SCHEDULED; version changed; scheduled_for_utc changed; due time not yet reached because of premature delivery. |
| Retryable | Transient DB/search/cache dependency failure before/after safe transaction boundaries, broker/network failure, lock timeout. |

| Contract item | Locked rule |
|---|---|
| Non-retry business failure | Current publish validation no longer passes: leave SCHEDULED state unchanged and dead-letter as BUSINESS_INVARIANT with diagnostics; do not publish partial/invalid content. |

### 9.1 Execution algorithm

1. Resolve site_id to Site Registry; if site is not ACTIVE, classify stale/no-op and never publish.
2. Open a site-scoped transaction and establish RLS tenant context.
3. Reload Post by site_id + post_id. Verify status=SCHEDULED, version==source_version and scheduled_for_utc exactly equals the payload schedule instant.
4. If now < scheduled_for_utc, release without mutation and re-delay to the authoritative instant; this is not an early publish.
5. Re-run DOC-07 publish eligibility/validation against current relationships/media/state.
6. Serialize the transition, apply WF-008 exactly once, increment version, create formal revision, clear schedule fields and set publication timestamps as DOC-07/08 require.
7. Commit, then enqueue CACHE_INVALIDATE, SEARCH_SYNC_POST and SITEMAP_BUILD_SITE/coalescing signals for the same site.
8. A redelivery after success reloads PUBLISHED state and becomes stale/no-op; no duplicate publish or duplicate revision is created.

## 10. Media-Processing Job Contract

| Contract item | Locked rule |
|---|---|
| Job type | JOB18-MEDIA-VARIANTS |
| Queue | q18.media |
| Payload | site_id, media_id, source_fingerprint=current original checksum identity, requested_variant_keys[], trigger |
| Variant allowlist | Exactly DOC-17 fixed catalog. Payload cannot request arbitrary dimensions/formats/crops. |
| Idempotency identity | media:{site_id}:{media_id}:{source_fingerprint}:{sorted_variant_keys_hash} |
| Serialization | One current generation per site/media; worker may skip already-verified requested keys. |
| Timeout/retry | R18-MEDIA: 15 minute attempt timeout, retry only transient/repairable failures. |
| Public baseline | native_webp semantics remain DOC-17. For normal upload/replace, it already exists before success; reconciliation may request native_webp repair if DOC-17 FAIL17-05 state is detected. |
| Completion | Persist only verified current-generation media_variants rows, then invalidate affected PublicMedia/PageData caches if availability changed. |

## 10.1 Stale-generation protection

- Worker reloads media under site scope and computes/reads the current generation identity. If it differs from source_fingerprint, the queued work is stale and must not write variants for the old generation.
- Each derivative object is written to the DOC-17 generation-specific private key before the corresponding media_variants row is inserted/updated.
- On retry, verified identical output may be reused; invalid partial objects are replaced before row commit.
- Canonical originals remain private. No retry path may use the original as a public fallback.

# 11. Incremental Public-Search Indexing

| Job | Trigger | Behavior |
|---|---|---|
| JOB18-SEARCH-SYNC-POST | Publish, Scheduled Publish, Update Published, Archive, Trash, Restore-to-nonpublic, Permanent Delete, published Revision Restore/bulk mutation; or fan-out from SEARCH-REFRESH-ENTITY/reconciliation | Reload Post under site scope; recompute current public eligibility and DOC-16 searchable projection. Upsert if eligible; delete if not. Payload action is only a hint. Payload requires the exact search source_fingerprint from Section 11.1. |
| JOB18-SEARCH-REFRESH-ENTITY | Category/Tag/Author rename/change affecting public indexed terms | Reload the same-site entity, verify its exact entity source_fingerprint from Section 11.1, resolve affected same-site Post IDs from authoritative relations, compute each Post search source_fingerprint and enqueue deduped SEARCH-SYNC-POST jobs. |

## 11.1 Exact search source-change identities

Search dedupe is keyed to the authoritative source change that can alter the DOC-16 public index projection. A Post version alone is insufficient because Category, Tag or Author display terms may change without incrementing posts.version. DOC-18 therefore uses a deterministic source_fingerprint; no database column is added.

| Identity | Exact construction |
|---|---|
| Existing Post search source_fingerprint | SHA-256 over canonical JSON v1 containing site_id, post_id, posts.version, posts.status, current author tuple (author_profile_id, display_name, status, updated_at) when present, current Category tuples sorted by category_id (category_id, name, status, updated_at), and current Tag tuples sorted by tag_id (tag_id, name, status, merged_into_tag_id, updated_at). Relationship membership is represented by the sorted tuple arrays. |
| Permanent-delete search source_fingerprint | Before terminal Post deletion, compute SHA-256 of UTF-8 string `search-delete-v1\|{site_id}\|{post_id}\|v{source_version}`. The Post ID is never reused; the worker may then safely delete the same-site index document if the row is gone. |
| Category entity source_fingerprint | SHA-256 over canonical JSON v1: entity_type=CATEGORY, category_id, name, status, updated_at. |
| Tag entity source_fingerprint | SHA-256 over canonical JSON v1: entity_type=TAG, tag_id, name, status, merged_into_tag_id, updated_at. |
| Author entity source_fingerprint | SHA-256 over canonical JSON v1: entity_type=AUTHOR, author_profile_id, display_name, status, updated_at. |

- Canonical JSON v1 means UTF-8, object keys in lexical order, arrays in the explicit order above, no insignificant whitespace, UUIDs in lowercase canonical text, and timestamps normalized to UTC RFC3339 with microsecond precision and `Z`. This is an internal identity encoding, not a new persistence format.
- JOB18-SEARCH-SYNC-POST dedupe identity is `search-sync:{site_id}:{post_id}:{source_fingerprint}`. Thus a later Post version or a changed related Category/Tag/Author search term cannot be suppressed by a prior 24-hour dedupe marker.
- JOB18-SEARCH-REFRESH-ENTITY dedupe identity is `search-refresh:{site_id}:{entity_type}:{entity_id}:{source_fingerprint}`. The RC1 placeholder `source-update` is prohibited.
- When SEARCH-REFRESH-ENTITY fans out, each affected Post gets a freshly computed current Post search source_fingerprint, so child sync dedupe reflects both current Post version and related entity terms.
- For an existing Post, source_version is still carried when available for diagnostics/stale comparison, but source_fingerprint is the dedupe generation identity.

### 11.2 Index write rules

- Every index document carries immutable site identity from the source record; a search worker never performs a network-wide query then partitions results afterward.
- DOC-16 owns fields, normalization/ranking/filtering and final ArticleSummary reprojection; DOC-18 only owns when/how the derived document converges.
- Incremental sync uses a serialization key of site_id + post_id so two updates to one Post cannot race into reversed index state.
- Before write, compare payload source_version/source_fingerprint to current authoritative source coordinates. If current truth is newer/different, do not write the stale snapshot; project current truth. A separately enqueued newer fingerprint is not suppressed by the older dedupe key.
- Index mutation is idempotent: repeated upsert of the same canonical projection or repeated delete is safe.

## 12. Full Site Reindex Contract

| Contract item | Locked rule |
|---|---|
| Job type | JOB18-SEARCH-REINDEX-SITE |
| Producer | API15-S-094 after search.view + search.reindex and durable 202 acceptance; internal lifecycle/recovery may also request same site-scoped job without creating a new user capability. |
| Queue | q18.search |
| Serialization | Exclusive site search-reindex lease; incremental search jobs for that site remain queued/blocked until activation completes. |
| Source | Current authoritative same-site PUBLISHED/public-eligible data only. |
| Build model | Build a shadow/site-scoped generation, validate tenant boundaries/count sanity, atomically activate, then release queued incremental work. Concrete search engine/index alias technology belongs DOC-23. |
| Status | Maintain DOC-16 READY/REINDEXING/DEGRADED semantics through RED18 search-status key. |
| Timeout/retry | R18-REINDEX: 30 minute attempt timeout; maximum 4 attempts. |

| Contract item | Locked rule |
|---|---|
| Failure | Never delete a known-valid active index merely because a rebuild fails. Mark DEGRADED when integrity/availability problem remains. |

**NO SECOND MANDATORY ADMIN INDEX**

The required rebuild target is the DOC-16 public derived index. Admin Post Search may use authoritative persistence or a later DOC-23-approved same-site acceleration structure; DOC-18 does not create a second product requirement.

# 13. XML Sitemap Generation Contract

| Contract item | Locked rule |
|---|---|
| Job type | JOB18-SITEMAP-BUILD-SITE |
| Queue | q18.sitemap |
| Granularity | Exactly one site per job; one build active per site. |
| Algorithm | Consume DOC-16 eligibility, lastmod, fixed resource names, type order, 25,000 URL shard limit and canonical URL ordering without modification. |
| Coalescing | Redis desired_seq/built_seq counters collapse multiple invalidations into at most one current build plus one follow-up when dirty state advances during a build. |
| Publication | Build a complete candidate generation, validate it, then atomically switch the site sitemap resource set. Do not publish half a generation. |
| Storage provider | Physical artifact store/serving adapter is an implementation choice under DOC-23/24; DOC-18 freezes generation/atomicity, not provider. |
| Timeout/retry | R18-SITEMAP: 15 minute attempt timeout; max 5 attempts. |

## 13.1 Sitemap invalidation triggers

- Publish/Update Published/Archive/Trash/Permanent Delete or published revision restore.
- Category/Tag/Author public change that affects archive eligibility, URL, lastmod or indexed content.
- SEO/index_policy/canonical_override changes that affect final sitemap eligibility.
- Site enable/disable/archive and canonical-domain changes.
- Approved static/Home public-affecting settings whose authoritative modification time/eligibility changes.
- Manual redirect edits do not create sitemap source URLs; DOC-16 live-route/canonical rules still decide inclusion.

# 14. Cache Invalidation Worker Contract

| Contract item | Locked rule |
|---|---|
| Job type | JOB18-CACHE-INVALIDATE |
| Queue | q18.cache |
| Payload target | Typed dependency identities (POST/CATEGORY/TAG/AUTHOR/MEDIA/NAVIGATION/SITE_SETTINGS/SITE/DOMAIN) and mode TARGETED or SITE_PUBLIC_ALL. Raw Redis keys/object-store keys are forbidden payload input. |
| Action | Invalidate owning-site Redis dependency keys; for DOC-17 events call the configured same-site media-CDN purge/revocation adapter. |
| Idempotency | Delete/purge/epoch advance operations are safe to repeat. |
| Timeout/retry | R18-CACHE: 60 second attempt timeout; max 8 attempts; high priority. |
| Security | Wrong-site/private/not-found negative responses are not inserted into long-lived shared cache. Admin/Preview bodies remain no-store. |

## 14.1 Invalidation modes

| Mode | When used | Mechanism |
|---|---|---|
| TARGETED | Normal post/taxonomy/author/media/navigation/settings changes | Resolve site-local reverse dependency sets and delete matching cache values; remove reverse-index entries. |
| SITE_PUBLIC_ALL | Site disable/archive, broad canonical-domain/site-public configuration change, recovery from dependency-index corruption | Atomically increment public cache epoch for the site, making all old public keys unreachable; targeted physical cleanup may follow asynchronously. |
| MEDIA_CDN | Replace/delete/public eligibility revoke/variant replacement | Derive purge namespace from validated site/media identity using DOC-17 delivery rules; never accept arbitrary purge URL or object key from job payload. |

# 15. Maintenance and Reconciliation Jobs

Maintenance jobs exist only to converge already-approved derived state from authoritative sources. They do not create user-visible features, new business states or destructive retention policy.

| Job type | Cadence / trigger | Checks and repairs |
|---|---|---|
| JOB18-SCHEDULE-RECONCILE-SITE | Every 60 seconds per ACTIVE site | Find due SCHEDULED posts from PostgreSQL and ensure deduped PUBLISH_DUE work exists. Does not publish directly without the normal publish job checks. |

| Job type | Cadence / trigger | Checks and repairs |
|---|---|---|
| JOB18-MEDIA-RECONCILE-SITE | Every 15 minutes per site with media activity or detected failure | Find ACTIVE media whose current generation lacks required native_webp/public baseline or known requested fixed variants; enqueue site/media/generation repair work. No deletion of old generations. |
| JOB18-SEARCH-RECONCILE-SITE | Every 15 minutes and after detected index integrity failure | Compare authoritative eligible Post identities/count/checks against same-site derived index; enqueue post syncs or site reindex when drift cannot be repaired incrementally. |
| JOB18-SITEMAP-RECONCILE-SITE | Every 15 minutes and after failed/pending dirty build | Compare desired_seq/built_seq and current site eligibility; enqueue build if dirty or generation missing/inconsistent. |

**DESTRUCTIVE CLEANUP DEFERRED**

DOC-17 marks superseded/orphan media objects as cleanup candidates, but retention/destructive purge policy is not frozen here. DOC-18 may detect/report candidates; actual retention/security/backup-safe deletion waits for the later authorities that own those policies.

# 16. Failure Classification and Retryability

| Class | Examples | Disposition |
|---|---|---|
| SUCCESS | Required side effect committed/verified | ACK; write success telemetry; clear dedupe/lease per contract. |
| STALE_NOOP | Rescheduled Post version mismatch; old media generation; entity now nonpublic; site no longer ACTIVE | ACK as terminal no-op. Do not DLQ ordinary expected staleness. |
| TRANSIENT | DB/network/broker/search/object-store/cache timeout, temporary 503, lease contention | Retry under family profile with backoff/jitter. |
| RATE_LIMITED | Downstream provider asks to retry later | Respect provider Retry-After when longer than computed backoff, bounded by profile cap. |
| BUSINESS_INVARIANT | Scheduled Post still SCHEDULED but current publish validation fails | Do not mutate; DLQ after first validated classification or configured family attempt; emit diagnostic. |
| SECURITY_INVARIANT | Missing site_id, target resolves to another tenant, malformed cross-site payload | Do not broaden scope; dead-letter immediately; security telemetry. |
| PERMANENT_DEPENDENCY | Unsupported fixed variant impossible under current validated source, corrupted authoritative reference requiring human repair | DLQ; do not loop indefinitely. |

# 17. Retry, Exponential Backoff and Attempt Timeouts

For retry number n starting at 1 after the first failed attempt, the base delay is `min(base_delay * 2^(n-1), max_delay)`. Add deterministic bounded jitter from 0% to 20% derived from the job idempotency hash so many same-time jobs do not retry in lockstep.

| Profile | Applies to | Max attempts | Base delay | Max delay | Attempt timeout |
|---|---|---|---|---|---|
| R18-PUBLISH | JOB18-PUBLISH-DUE | 8 | 15 s | 5 min | 60 s |
| R18-MEDIA | JOB18-MEDIA-VARIANTS | 6 | 30 s | 15 min | 15 min |
| R18-SEARCH | SEARCH_SYNC/ REFRESH | 6 | 5 s | 2 min | 60 s |
| R18-REINDEX | SEARCH_REINDEX_SITE | 4 | 60 s | 15 min | 30 min |
| R18-SITEMAP | SITEMAP_BUILD_SITE | 5 | 30 s | 10 min | 15 min |
| R18-CACHE | CACHE_INVALIDATE | 8 | 2 s | 60 s | 60 s |
| R18-MAINT | maintenance/ reconcile jobs | 3 | 60 s | 10 min | 10 min |

- Timeout means the worker must stop/abort the attempt or lose its execution lease; it does not authorize partial continuation in a detached thread/process.
- A timed-out attempt can be retried only when the job family side effects are idempotent or fenced by current version/generation/state.
- Later PDF-20 may size worker pools/throughput and set stricter operational SLOs, but it must not weaken these correctness-oriented attempt/retry semantics without reopening DOC-18.

# 18. Idempotency and Deduplication

## 18.1 Distinguish API idempotency from worker idempotency

| Layer | Purpose | Identity |
|---|---|---|
| DOC-15 Idempotency-Key | Prevent ambiguous duplicate HTTP create/command execution for 24 hours. | Actor/scope/method/path/body fingerprint. |
| DOC-18 job idempotency | Collapse repeated internal delivery/trigger/reconciliation for one intended derived effect. | Site + job type + entity/version/generation/desired sequence as defined per job. |

## 18.2 Dedupe rules

- Each intended job effect has one RED18-DEDUPE key derived from the exact Appendix A identity. The Redis value is a small state record; a bare Boolean marker is not sufficient.
- Dedupe TTL for an ENQUEUED or TERMINAL ordinary job is 24 hours after terminal completion or the maximum retry horizon, whichever is longer. Delayed publish retention is at least through `scheduled_for_utc + 24 hours`.

- Reconciliation may suppress equivalent enqueue only when a current ENQUEUED/TERMINAL record proves the exact same identity is pending/completed, or while a short live CLAIMED record is still inside its 60-second enqueue grace. CLAIMED never proves durable queue acceptance.
- Dedupe never merges different sites, search source_fingerprints, Post versions where they are the source generation, media generations, or sitemap desired sequences.
- Loss of Redis dedupe state may cause duplicate enqueue but must never cause a missed authoritative effect; at-least-once delivery, source-state revalidation and leases/fences make duplicates safe.

## 18.3 Enqueue-claim state machine and crash recovery

| State | How entered | Meaning / recovery rule |
|---|---|---|
| CLAIMED | Producer atomically creates RED18-DEDUPE with SET-NX, random claim_token and exact 60-second TTL before attempting broker enqueue. | Provisional ownership of the enqueue attempt only. It is NOT proof a message exists. Producer must stop waiting for broker acceptance before the claim expires. |
| ENQUEUED | Only after positive durable broker acceptance, producer CAS-updates the same record using matching claim_token; or a worker receiving the message repairs CLAIMED/missing state to ENQUEUED. | Proof that the broker accepted/has delivered this exact identity. Set the job-specific retention horizon. Retries/redeliveries retain the same identity. |
| TERMINAL | Worker finishes SUCCESS or STALE_NOOP, or moves the job to DLQ after final disposition; atomically record terminal outcome. | Equivalent same-identity reconciliation is suppressed for the retention horizon. A new source version/fingerprint/generation/desired_seq uses a different key and is not suppressed. |

1. Compute exact idempotency_identity and RED18-DEDUPE key.
2. Create CLAIMED with claim_token via SET-NX and 60-second TTL. If another live exact-identity record exists, do not enqueue a second copy.
3. Attempt durable queue acceptance. If the broker positively rejects/fails and durable acceptance is known not to have occurred, compare-and-delete CLAIMED using claim_token immediately.
4. If producer outcome is ambiguous or the producer crashes before confirmation, do not manufacture ENQUEUED. CLAIMED expires automatically after 60 seconds; reconciliation may then reclaim and enqueue. If the broker actually accepted before the crash, duplicate delivery is allowed and must be harmless.
5. After positive durable acceptance, CAS CLAIMED -> ENQUEUED using the same claim_token and set the normal retention horizon. A worker receipt may repair a still-CLAIMED or missing record to ENQUEUED because receipt itself proves queue existence.
6. On terminal worker outcome, write TERMINAL with outcome and terminal_at. Queue/DLQ remains the delivery record; Redis remains derived and may be rebuilt/lost without changing canonical business truth.

**RELIABILITY INVARIANT**

A CLAIMED dedupe record can delay a duplicate enqueue for at most 60 seconds; it can never permanently convince reconciliation that a queue message exists. Only durable broker acceptance or actual worker receipt can establish ENQUEUED. This closes the post-commit enqueue-failure hole without adding a DOC-08 outbox/jobs table.

# 19. Concurrency, Serialization, Leases and Stale Workers

| Work family | Serialization key | Rule |
|---|---|---|
| Scheduled publish | site_id + post_id | One publish transition at a time; PostgreSQL row/version checks remain final authority. |
| Media variants | site_id + media_id + generation | One generation coordinator; variant writes may parallelize only if each variant_key is independently fenced and current generation is rechecked before row commit. |
| Search post sync | site_id + post_id | Latest canonical projection wins; older queued version cannot overwrite newer state. |
| Full reindex | site_id + search-reindex | Exclusive with same-site incremental index activation; incremental messages wait then catch up. |
| Sitemap build | site_id | One active build; dirty sequence can advance during build and force one follow-up. |
| Cache invalidation | site_id + dependency/mode | May coalesce; broad epoch advance serializes per site. |
| Maintenance | site_id + maintenance-family | Prevent overlapping same-site full scans. |

## 19.1 Lease mechanics

- Execution lease TTL = attempt timeout + 60 seconds. Long-running workers renew at least every one-third of the attempt timeout.
- Lease ownership uses a unique attempt token. Releasing a lease requires matching the token; a stale worker must not delete a newer worker lease.
- Where the derived target supports fencing/generation checks, commit is conditioned on the current generation/version. Redis lock ownership alone is never the only correctness barrier for PostgreSQL mutations.
- If a lease expires and a second worker starts, the first worker must fail its next fence/current-state check and stop before committing stale output.

# 20. Dead-Letter Queues and Replay

| Source queue | DLQ |
|---|---|
| q18.publish | dlq18.publish |
| q18.media | dlq18.media |
| q18.search | dlq18.search |
| q18.sitemap | dlq18.sitemap |
| q18.cache | dlq18.cache |
| q18.maintenance | dlq18.maintenance |

## 20.1 DLQ record

- Preserve original validated envelope, queue name, final error_class/error_code, attempt_count, first_failed_at, last_failed_at and correlation_id.
- Do not include raw uploaded bytes, article body content, authentication tokens, signed URLs, Redis secrets, object-store credentials or private object keys beyond the already-internal identifiers strictly needed to reproduce the job.
- Expected stale/no-op outcomes do not enter DLQ.
- SECURITY_INVARIANT may dead-letter immediately without normal retry.

## 20.2 Replay

1. Replay is an internal operational action, not a new Central Admin page/API/capability.
2. Preserve the original job_type/site_id/idempotency identity; create a fresh attempt token and correlation child ID.
3. Re-run all Site Registry, RLS, ownership, current-state/version/generation and eligibility checks. Replay is never “force execute”.
4. If authoritative state makes the job stale, classify stale/no-op and close it.
5. Never replay a site job under GLOBAL context or another tenant because its target exists there.

# 21. Redis Namespace and Exact Key Grammar

**NAMESPACE RULE**

Redis database numbers are not tenant isolation. Every site-owned key MUST begin with the explicit internal tenant prefix `bp:v1:s:{site_id}:`. Global keys are confined to `bp:v1:g:` and may be used only by explicit CTX-GLOBAL coordinator paths.

| Key ID | Pattern | Meaning |
|---|---|---|
| RED18-EPOCH | bp:v1:s:{site_id}:epoch:public | Integer site public-cache epoch. |
| RED18-PAGE | bp:v1:s:{site_id}:cache:page:e{epoch}:{host_hash}:{route_hash} | Canonical public page/render cache when used. |
| RED18-DATA | bp:v1:s:{site_id}:cache:data:e{epoch}:{page_id}:{identity_hash}:{query_hash} | Public PageData/projection cache. |
| RED18-OBJ | bp:v1:s:{site_id}:cache:obj:{entity_type}:{entity_id}:v{version_token} | Site-owned derived object/projection cache. |
| RED18-DEP | bp:v1:s:{site_id}:dep:{dep_type}:{dep_id} | SET of current cache keys depending on one site-owned identity. |
| RED18-DEDUPE | bp:v1:s:{site_id}:dedupe:{job_type}:{identity_hash} | Worker dedupe state record: CLAIMED -> ENQUEUED -> TERMINAL. CLAIMED has exact 60s TTL and is never durable-acceptance proof. |
| RED18-LOCK | bp:v1:s:{site_id}:lock:{purpose}:{identity_hash} | Execution lease with attempt token. |
| RED18-SEARCH-STATUS | bp:v1:s:{site_id}:search:status | DOC-16 READY/REINDEXING/DEGRADED coordination state. |
| RED18-SMAP-DESIRED | bp:v1:s:{site_id}:sitemap:desired_seq | Monotonic dirty sequence. |
| RED18-SMAP-BUILT | bp:v1:s:{site_id}:sitemap:built_seq | Latest atomically published sequence. |
| RED18-G-LOCK | bp:v1:g:lock:{purpose} | Explicit global scheduler/maintenance coordinator lease only. |

### 21.1 Hashing rules

- host_hash, route_hash, identity_hash and query_hash use lowercase hexadecimal SHA-256 truncated to the first 32 hex characters for key compactness; the full canonical input remains reconstructable from authoritative data/log correlation, not from the hash alone.
- Canonical host/path/query input is normalized by DOC-09/DOC-15 before hashing. Two semantically identical normalized requests produce the same key.
- Hashing is collision reduction/compactness, not security or authorization.

## 22. Redis Cache Classes, Values and TTLs

| Class | TTL / expiry | Value rules |
|---|---|---|
| Public PageData/projection | 300 seconds | Contains only canonical safe public projection; includes schema_version, site_id assertion, cached_at and source version tokens. Must be read only after trusted SiteContext/site-lifecycle resolution. |
| Public page/render cache (if implementation uses it) | 300 seconds | Key includes canonical host + route; never shared by path alone across domains. No Preview/Admin content. |
| Site-owned object/projection cache | 600 seconds | Key includes entity identity + source version token; stale version keys become unreachable when callers request current version. |
| Dependency set RED18-DEP | 900 seconds after latest member refresh | Must outlive dependent values by at least 300 seconds. Members are same-site cache keys only. |
| Search status | No automatic TTL while valid | Derived state; missing/corrupt status is reconstructed from search adapter or reported DEGRADED, never guessed READY. |
| Sitemap desired/built sequence | No automatic TTL while site exists | Derived coordination counters; loss triggers reconcile from authoritative site state. |
| Dedupe | Per Section 18 | At least retry horizon; delayed publish through due+24h. |
| Execution lease | attempt timeout + 60 seconds | Renewed; token/fence protected. |

### 22.1 Prohibited Redis content

- No Admin/Preview response-body cache that bypasses DOC-15 no-store.
- No raw password/MFA/session/preview credential, API secret, signed private media URL or object-store credential.
- No all-site cache entry later filtered by the browser/frontend.
- No wrong-site/private negative cache with long lifetime.
- No Redis key keyed only by slug/path/media_id/post_id when the value varies by tenant.

## 23. Dependency Tracking and Invalidation Rules

A cache entry that depends on site-owned entities registers same-site dependency membership in RED18-DEP sets. The worker resolves dependencies; producers do not submit arbitrary Redis key strings.

| Mutation/effect | Dependencies invalidated |
|---|---|
| Publish / Update Published / Scheduled Publish / published revision restore | Post; Article route; Home/collection projections that include it; affected Category/Tag/Author archive projections; related-content projections where upstream relation says applicable. |
| Archive / Trash / Permanent Delete | Same as above, plus remove current public eligibility caches. |
| Category/Tag/Author edit | Entity archive/detail + post/search/public projections that consume the changed entity metadata. |
| Navigation/Home controls/site public settings | Home/navigation/site-settings dependent PageData for that site. |
| SEO index/canonical changes | Affected page projection + sitemap dirty sequence; search only when DOC-16 searchable projection/destination changes. |
| Media metadata/variant availability | All PageData dependencies registered for that media_id; binary CDN purge only when DOC-17 requires it. |
| Media replace/delete/eligibility revoke | Media dependency cache + required DOC-17 CDN purge/revocation. |
| Site disable/archive | SITE_PUBLIC_ALL epoch advance; search/sitemap lifecycle jobs; no public cache hit may bypass current lifecycle. |
| Canonical-domain change | SITE_PUBLIC_ALL epoch advance; sitemap regeneration; search destination reprojection/reindex as DOC-16 requires. |

# 24. Search-Status Coordination in Redis

| Field in RED18-SEARCH-STATUS | Rule |
|---|---|
| state | Exactly READY \| REINDEXING \| DEGRADED from DOC-16. |
| reindex_started_at | Present while/after latest accepted full rebuild attempt. |
| last_indexed_at | Most recent successful full or authoritative incremental synchronization instant known to the search service. |
| active_generation | Internal opaque derived-index generation token; never exposed as public/Admin contract field. |
| message_code | Internal safe diagnostic code mapped to DOC-15 SearchStatusResponse.message without engine secrets. |
| updated_at | Worker status-write instant. |

## 24.1 State transitions

```
READY --accepted reindex--> REINDEXING
REINDEXING --validated atomic activation--> READY
REINDEXING --failed rebuild / integrity issue--> DEGRADED
DEGRADED --successful repair/reindex--> READY

Missing Redis status != READY by assumption.
The service must inspect the active search adapter/index state or return DEGRADED.
```

# 25. Sitemap Dirty/Build Coordination

1. Any sitemap-affecting event atomically increments `desired_seq` for that site and attempts to enqueue one deduped SITEMAP_BUILD_SITE.
2. Worker acquires site sitemap lease and snapshots target_seq = desired_seq.
3. Build complete DOC-16 resource set from current authoritative truth, not from event payload deltas.
4. Validate no mixed tenant URLs, fixed resource order, shard size/order and canonical domain.
5. Publish candidate generation atomically, then set built_seq = target_seq.
6. If desired_seq > built_seq because another event arrived during the build, enqueue one follow-up build. If equal, stop.
7. Retry/replay never publishes a candidate built for an old canonical domain or inactive site without revalidation.

# 26. Queue Ordering, Priority and Fairness

| Priority | Work |
|---|---|
| P0 | SITE_PUBLIC_ALL invalidation, media delete/revocation purge, security/lifecycle cache invalidation. |
| P1 | Due scheduled publication and its immediate post-commit invalidation/search/sitemap signals. |
| P2 | Media native_webp repair, incremental search sync, targeted cache invalidation. |
| P3 | Normal remaining media variants, sitemap build. |
| P4 | Full reindex and maintenance/reconciliation scans. |

- Priority influences dispatch order only; it cannot bypass tenant locks, state validation or required retries.
- Within a priority class, the queue should avoid one site monopolizing all capacity. Concrete worker-count/fairness tuning belongs PDF-20/23, but starvation of other sites is prohibited.
- FIFO is not relied upon for correctness. Version/generation/state checks make out-of-order delivery safe.

# 27. Explicit Global Fan-Out and Site Enumeration

**GLOBAL IS DELIBERATE**

A scheduler/maintenance coordinator may operate under the frozen CTX-GLOBAL context only to read the global Site Registry and enqueue one site-scoped job per eligible site. It does not open a tenant-blind repository for Posts/Media/Search documents.

1. Acquire RED18-G-LOCK for the coordinator purpose so duplicate schedulers do not create uncontrolled fan-out.
2. Read Site Registry through explicit global path.
3. For each eligible site, create a site-scoped envelope with that site_id and enqueue the relevant reconcile job.
4. Each worker independently resolves CTX-WORKER-SITE and RLS before reading site-owned rows.
5. Coordinator failure midway is safe: next run repeats enumeration; site-job dedupe prevents duplicate harmful effects.

# 28. Site Lifecycle and Domain-Change Background Effects

| Upstream event | Required DOC-18 effects |
|---|---|
| Site ENABLED | Advance/establish site public epoch; reconcile public search; generate current sitemap; normal requests still depend on DOC-09/DOC-04 readiness. |
| Site DISABLED | Immediately make Redis public cache entries ineligible via epoch advance; enqueue search ineligibility/reconciliation; suppress/regenerate sitemap to no ordinary public output; site-owned worker jobs revalidate and no-op if effect is no longer legal. |
| Site ARCHIVED / deletion-request lifecycle | Same fail-closed derived behavior as upstream lifecycle requires; no destructive data purge is invented. |
| Canonical domain changed | Advance public epoch; rebuild sitemap on new canonical host; refresh/reindex search destinations according to DOC-16; never change site_id ownership. |
| Alias changed | Host-resolution cache if any must be invalidated; canonical site-owned derived state remains under same site_id. |

# 29. Failure and Recovery State Matrix

| Failure | Required recovery |
|---|---|
| Queue unavailable after Post schedule commit | Schedule remains authoritative; scheduler reconcile within 60s detects due work and enqueues when queue returns. |
| Queue unavailable before API15-S-094 202 | Do not claim accepted; existing DOC-15 503 class. |
| Worker crashes mid-publish before DB commit | Lease expires; redelivery re-runs state/version checks; no committed transition means safe retry. |
| Worker crashes after publish commit before downstream enqueue | Post remains PUBLISHED; cache TTL and search/sitemap reconciliation converge; next duplicate publish job is stale no-op. |
| Redis cache loss | Authoritative DB unaffected; cache cold start; public requests rebuild safe projections after SiteContext; dedupe/status/sitemap coordination reconstructed by queues/reconcilers. |
| Search index lost/corrupt | Status DEGRADED; site reindex reconstructs from authoritative public-eligible data. |
| Sitemap artifact generation fails | Keep last validated generation only if still safe/current-site eligible; otherwise resource may be unavailable until successful rebuild; never publish partial mixed data. |
| Media non-baseline variant fails | Keep verified native_webp/public safe variants; retry missing fixed variants; never expose private original. |
| Media old-generation job executes after replace | Generation fingerprint mismatch -> stale no-op. |
| CDN purge adapter temporary failure | Retry high-priority cache job; origin/public eligibility still denies invalid media on misses; emit high-priority operational signal per DOC-17. |

| Failure | Required recovery |
|---|---|
| DLQ replay after entity changed site? | Ordinary site-owned entity cannot move site; mismatch is security invariant, not a reason to adopt another tenant. |

# 30. Logging, Metrics and Correlation

## 30.1 Structured worker events

| Event | When emitted |
|---|---|
| job.enqueued | Producer/maintenance accepted one logical job. |
| job.started | Worker acquired execution lease/context. |
| job.succeeded | Required side effect committed/verified. |
| job.stale_noop | Current authoritative state made work unnecessary/obsolete. |
| job.retry_scheduled | Transient attempt failed and next retry was scheduled. |
| job.dead_lettered | Retry budget exhausted or immediate permanent/security classification. |
| job.replayed | Internal operational replay began after full revalidation. |
| cache.invalidated | Targeted deletion/epoch advance/CDN purge outcome. |
| search.reindex_state | READY/REINDEXING/DEGRADED transition. |
| sitemap.published | Validated site generation atomically activated. |
| media.variant_generated | Verified current-generation derivative committed. |

## 30.2 Required log fields

- UTC timestamp, environment, queue, job_type, site_id for site jobs, entity_type/entity_id when applicable, attempt number, duration_ms, correlation_id, causation_id, outcome/error_class/error_code.
- Log an idempotency hash, not raw client Idempotency-Key values.
- Do not log ArticleBody content, uploaded bytes, credentials, signed URLs, object-store secrets, session/Preview tokens or raw Redis passwords.

## 30.3 Metric families

| Metric | Dimensions |
|---|---|
| worker_job_total | queue, job_type, outcome |
| worker_job_duration_seconds | queue, job_type |
| worker_retry_total | job_type, error_class |
| worker_dlq_depth | queue |
| worker_queue_depth | queue |
| worker_oldest_job_age_seconds | queue |

| Metric | Dimensions |
|---|---|
| worker_lock_contention_total | purpose |
| cache_invalidation_total | mode, outcome |
| search_reindex_total | outcome |
| sitemap_build_total | outcome |
| media_variant_job_total | outcome |

Alert thresholds, capacity targets, tracing backend, retention and SLOs are PDF-20/19 authority. DOC-18 freezes the event/metric meaning needed for later observability.

# 31. Security and Privacy Boundaries

- Workers use dedicated runtime credentials with only the infrastructure/database privileges needed for their scoped function; exact secret management and credential rotation are PDF-19.
- Normal site workers have no implicit BYPASSRLS/superuser authority. SET LOCAL tenant context is established inside each DB transaction.
- Queue payloads never contain user passwords, MFA secrets, Preview tokens, raw object-store credentials or public authorization tokens.
- Job payload validation rejects unknown job_type/schema versions and unknown properties before target access.
- Redis is not a security boundary that can authorize site access. A cache hit is used only after trusted request/site context has been resolved.
- No worker exposes queue/DLQ/cache inspection as a new Admin module or API endpoint.
- Global maintenance paths do not convert site-owned rows into global ownership.

# 32. End-to-End Execution Examples

## 32.1 Schedule -> due publish -> derived convergence

```
1) API15-S-017 commits Post(site=A, status=SCHEDULED, version=42, scheduled_for_utc=T)
2) Producer enqueues JOB18-PUBLISH-DUE(site=A, post=P, source_version=42, not_before=T)
3) At T worker reloads A/P under RLS; verifies ACTIVE + SCHEDULED + v42 + due
4) Worker executes DOC-07 WF-008 -> PUBLISHED v43; schedule cleared; revision committed
5) After commit enqueue:
   - JOB18-CACHE-INVALIDATE(site=A, POST=P)
   - JOB18-SEARCH-SYNC-POST(site=A, POST=P, source_fingerprint=current-search-source)
   - sitemap desired_seq++ and JOB18-SITEMAP-BUILD-SITE(site=A)
6) Any duplicate publish message sees PUBLISHED/v43 and ACKs STALE_NOOP.
```

## 32.2 Media replace -> remaining variants + purge

```
1) DOC-17 replace validates/stages private original + sanitized native_webp
2) Atomic current-generation switch commits under same site/media_id
3) Enqueue JOB18-MEDIA-VARIANTS with new generation fingerprint and missing fixed keys
4) Enqueue JOB18-CACHE-INVALIDATE for MEDIA media_id + required old-generation CDN purge
5) Old-generation media job later arrives -> fingerprint mismatch -> STALE_NOOP
6) PublicMedia continues to use only verified current sanitized derivatives.
```

### 32.3 Reindex command

```
1) API15-S-094 authorizes search.view + search.reindex and receives Idempotency-Key
2) API returns 202 only after durable site-scoped JOB18-SEARCH-REINDEX-SITE acceptance
3) Worker sets search status REINDEXING, builds shadow same-site public index from current truth
4) Validate tenant boundary, atomically activate generation, set READY + last_indexed_at
5) Queued incremental SEARCH_SYNC_POST jobs resume and catch up
6) Failure leaves safe prior generation if valid and sets DEGRADED.
```

## 33. Cross-Document Consistency Audit

### 33.1 RC1 controlled-issue reconciliation

| RC1 audit item | RC2 resolution | Result |
|---|---|---|
| Search-sync dedupe used site+post only | Section 11.1 + Appendix A now use site+post+exact search source_fingerprint; related entity term changes produce a new fingerprint even when posts.version is unchanged. | RESOLVED |
| SEARCH-REFRESH-ENTITY used undefined source-update | Section 11.1 defines exact Category/Tag/Author source_fingerprint encodings and prohibits the RC1 placeholder. | RESOLVED |
| Dedupe claim could survive failed enqueue and block reconciliation | Section 18.3 defines CLAIMED/ENQUEUED/TERMINAL, exact 60s provisional claim TTL, CAS release/promotion and ambiguous-crash recovery. | RESOLVED |
| CTX-GLOBAL-SYSTEM renamed an upstream context | Sections 4/8/21/27 now use the frozen DOC-04 CTX-GLOBAL terminology; no new trusted context class exists. | RESOLVED |

### 33.2 Upstream authority audit

| Authority | Result | Audit conclusion |
|---|---|---|
| DOC-01 | PASS | Shared worker/cache infrastructure preserved; exact site-aware keys/job mechanics defined here without per-site backend clones. |
| DOC-02 | PASS | No permission added; workers do not treat payloads as user authority; reindex authorization remains DOC-02/15. |
| DOC-03 | PASS | Only existing scheduled publish/media/search/reindex/system effects implemented; no generic automation capability. |
| DOC-04 | PASS | Uses frozen CTX-GLOBAL only for deliberate Site Registry enumeration, then CTX-WORKER-SITE for every tenant job; fail-closed scope and site-aware retries/DLQ/cache/search/sitemap enforced. |

| Authority | Result | Audit conclusion |
|---|---|---|
| DOC-05/06 | PASS | No new Public/Admin Page ID created for queues/DLQ/maintenance. |
| DOC-07 | PASS | Five-state Post workflow unchanged; schedule generation implemented using existing version+scheduled_for tuple; stale/early publish rules preserved. |
| DOC-08 | PASS | No jobs/outbox/cache table added; workers use existing rows/RLS/version fields; Redis remains derived. |
| DOC-09 | PASS | Workers consume current canonical routing/domain state; do not redefine routes/redirect precedence. |
| DOC-10/11 | PASS | Reindex/public failure semantics preserved; no hidden worker UI. |
| DOC-12/13 | PASS | No component/data schema redefinition; public PageData never exposes internal job/cache fields. |
| DOC-14 | PASS | No frontend imports/runtime state from Redis/queues; 14 independent apps unchanged. |
| DOC-15 RC6 | PASS | 101-operation API contract unchanged; 202 reindex acceptance integrated without adding endpoint/status/body; Admin/Preview no-store preserved. |
| DOC-16 RC4 | PASS | Ranking/search fields/sitemap resources/eligibility/sharding/redirect rules unchanged; DOC-18 only executes rebuild/invalidation. |
| DOC-17 RC2 | PASS | Private original/sanitized native_webp/16-key catalog unchanged; only asynchronous remaining/repair work and cache/CDN purge mechanics defined. |

**AUDIT RESULT**

RC2 closes all four issues raised against RC1 without changing upstream product/API/database semantics. After correction, no unresolved upstream contradiction or blocking missing dependency remains for DOC-18 scope. Two deliberate later dependencies remain non-blocking: concrete queue/search/sitemap storage adapters and process/runtime packages belong DOC-23/24; security/observability retention and performance/DR thresholds belong DOC-19/20.

# 34. Traceability Registry

| Trace ID | Source | DOC-18 implementation lock | Home |
|---|---|---|---|
| TR18-001 | DOC-01 Cache/Workers direction | Shared site-aware cache/worker system | Sections 3,21-23 |

| Trace ID | Source | DOC-18 implementation lock | Home |
|---|---|---|---|
| TR18-002 | DOC-04 CTX-WORKER-SITE | Payload+revalidation+RLS worker context | Sections 4-5,19-20 |
| TR18-003 | DOC-04 cache isolation | Tenant/host-aware Redis keys and invalidation | Sections 21-23 |
| TR18-004 | DOC-07 Schedule/Reschedule | Delayed publish + stale generation tuple | Sections 8-9 |
| TR18-005 | DOC-07 downstream public effects | Cache/search/sitemap jobs after public mutations | Sections 7,11,13-14 |
| TR18-006 | DOC-08 no queue/cache tables | Derived Redis/queue state only | Sections 7,21 |
| TR18-007 | DOC-08 post version/schedule fields | Scheduled publish current-state fencing | Sections 8-9 |
| TR18-008 | DOC-10 FC-SEARCH-07 | Authorized reindex accepted then worker executes | Section 12 |
| TR18-009 | DOC-15 API15-S-094 | 202 acceptance without public job ID | Sections 6-7,12 |
| TR18-010 | DOC-16 indexing lifecycle | Incremental sync/reindex convergence | Sections 11-12,24 |
| TR18-011 | DOC-16 sitemap rules | Dirty/build coalescing + atomic generation | Sections 13,25 |
| TR18-012 | DOC-17 media processing | Remaining/repair variant jobs with generation fingerprint | Section 10 |
| TR18-013 | DOC-17 cache/CDN invalidation | Site/media-derived purge adapter | Sections 14,23 |
| TR18-014 | DOC-13 no internal fields | No job/cache leakage into PageData | Sections 5,31 |
| TR18-015 | DOC-14 frontend isolation | No frontend direct queue/Redis dependency | Sections 3,31 |
| TR18-016 | DOC-16 public index source terms | Exact search source_fingerprint changes with Post version or related Category/Tag/Author indexed terms | Section 11.1, Appendix A |
| TR18-017 | DOC-04 CTX-GLOBAL | Global coordinator only enumerates Site Registry; tenant processing always re-enters CTX-WORKER-SITE | Sections 4,27 |
| TR18-018 | DOC-08 no outbox/jobs table | Redis CLAIMED/ENQUEUED/TERMINAL enqueue recovery without new persistence | Sections 7,18.3 |

# 35. Comprehensive Acceptance Criteria

| QA ID | Area | Release-blocking assertion |
|---|---|---|
| QA18-001 | Context | Every site-owned job contains site_id and fails closed if one trusted site cannot be reconstructed. |

| QA ID | Area | Release-blocking assertion |
|---|---|---|
| QA18-002 | Context | Worker reloads target under CTX-WORKER-SITE and verifies target.site_id before side effects. |
| QA18-003 | Context | Worker DB transactions establish transaction-local tenant/RLS context; pooled connections do not retain prior tenant. |
| QA18-004 | Queue | Exactly six logical queues from Section 6 are used; provider-specific names may map 1:1 but not merge tenant semantics. |
| QA18-005 | Queue | Delivery is treated as at-least-once and no correctness claim depends on FIFO/exactly-once. |
| QA18-006 | Queue | Message is ACKed only after side effect commits or a terminal stale/no-op classification. |
| QA18-007 | Queue | 202 reindex is returned only after durable queue acceptance; worker/job IDs remain absent from API response. |
| QA18-008 | Producer | No worker observes uncommitted source mutation as intended truth. |
| QA18-009 | Producer | Post-commit enqueue failure does not invent compensating content state; recovery path is observable and reconcilable. |
| QA18-010 | Schedule | Schedule generation uses existing site_id/post_id/version/scheduled_for_utc tuple; no new DB field/table. |
| QA18-011 | Schedule | Reschedule makes earlier job stale because current version/schedule instant no longer matches. |
| QA18-012 | Schedule | Manual early publish/unschedule/archive causes pending delayed job to no-op rather than publish again. |
| QA18-013 | Schedule | Schedule reconcile runs at least every 60 seconds per ACTIVE site and recovers missing due jobs. |
| QA18-014 | Schedule | Worker never converts local schedule time; due comparison uses persisted UTC instant. |
| QA18-015 | Publish | Scheduled worker revalidates ACTIVE site, same-site Post, SCHEDULED state, version, due time and publish eligibility. |
| QA18-016 | Publish | Duplicate/redelivered scheduled-publish work never creates a second publish transition/revision. |
| QA18-017 | Media | Media job payload includes same-site media_id + current generation fingerprint + allowlisted fixed variant keys. |
| QA18-018 | Media | Old-generation media job cannot write after replacement. |
| QA18-019 | Media | No media worker exposes private canonical original as public fallback. |

| QA ID | Area | Release-blocking assertion |
|---|---|---|
| QA18-020 | Media | Variant row is committed only after current-generation derivative object exists and verifies. |
| QA18-021 | Search | Incremental index sync recomputes eligibility from current PostgreSQL truth; payload UPSERT/DELETE hint is not authority. |
| QA18-022 | Search | Same-site Post sync serializes by site_id+post_id and newer state cannot be overwritten by older queued work. |
| QA18-023 | Search | Full reindex rebuilds exactly one site public index and does not create network-wide public search. |
| QA18-024 | Search | Full reindex status uses only READY/REINDEXING/DEGRADED meanings from DOC-16. |
| QA18-025 | Search | Incremental jobs are not lost across full reindex activation; they resume/catch up after site-wide reindex lease releases. |
| QA18-026 | Sitemap | One site sitemap build never emits another site canonical domain/URL. |
| QA18-027 | Sitemap | Build applies DOC-16 eligibility/lastmod/sharding/order exactly and publishes atomically. |
| QA18-028 | Sitemap | desired_seq/built_seq coalescing schedules a follow-up when events arrive during a build. |
| QA18-029 | Cache | Every site-owned Redis key starts with bp:v1:s:{site_id}: and no key uses slug/path/entity ID alone for tenant-varying values. |
| QA18-030 | Cache | Public PageData/page caches expire after 300 seconds and targeted invalidation removes known dependencies sooner. |
| QA18-031 | Cache | Site disable/archive advances site public epoch so old Redis public cache keys become unreachable. |
| QA18-032 | Cache | Admin/Preview response bodies are not placed in shared Redis response cache. |
| QA18-033 | Cache | Wrong-site/private/not-found responses are not stored as long-lived shared negative entries. |
| QA18-034 | Cache | Media purge job derives site/media purge target from authoritative DOC-17 identity and never accepts arbitrary raw object key. |
| QA18-035 | Retry | Each job family uses exactly the Section 17 retry profile or a formally revised DOC-18 profile. |
| QA18-036 | Retry | Backoff is capped exponential with deterministic 0-20% jitter; Retry-After may extend but not bypass max-attempt policy. |
| QA18-037 | Retry | Timed-out attempts cannot continue committing after lease/fence loss. |

| QA ID | Area | Release-blocking assertion |
|---|---|---|
| QA18-038 | Idempotency | Worker dedupe identity always includes site and the exact job-specific source generation: search source_fingerprint, Post version/due tuple, media generation, or sitemap desired_seq as applicable. |
| QA18-039 | Idempotency | Delayed publish dedupe survives through due time + 24h even for far-future schedules. |
| QA18-040 | Idempotency | DOC-15 Idempotency-Key and DOC-18 job dedupe are not conflated. |
| QA18-041 | Concurrency | PostgreSQL current version/state remains final fence for scheduled publish; Redis lock alone is insufficient. |
| QA18-042 | Concurrency | Releasing a Redis lease requires matching attempt token; stale attempt cannot release a newer lease. |
| QA18-043 | DLQ | Each source queue has a corresponding DLQ; expected stale/no-op jobs do not enter DLQ. |
| QA18-044 | DLQ | DLQ replay preserves site/job identity and re-runs current ownership/lifecycle/state checks. |
| QA18-045 | DLQ | Security invariant mismatch dead-letters immediately without cross-site lookup/broadening. |
| QA18-046 | Maintenance | Global scheduler enumerates Site Registry explicitly and then emits site-scoped jobs; null site never means all sites. |
| QA18-047 | Maintenance | Maintenance jobs repair derived state only and do not create new product states or destructive retention policy. |
| QA18-048 | Recovery | Redis loss cannot change canonical editorial state; cache/search/sitemap coordination is rebuilt from authoritative truth. |
| QA18-049 | Recovery | Search index loss/corruption leads to DEGRADED + site rebuild, not cross-site fallback data. |
| QA18-050 | Recovery | Failed sitemap build never publishes half/mixed generation. |
| QA18-051 | Recovery | Cache/CDN purge temporary failure retries and remains observable without exposing private original or wrong-site content. |
| QA18-052 | Observability | Structured logs include queue/job/site/correlation/attempt/outcome and omit secrets/raw user content. |
| QA18-053 | Observability | Required worker metrics exist with bounded dimensions; thresholds remain later PDF-20 authority. |
| QA18-054 | Security | No frontend or public API client can access Redis/queue/DLQ contracts directly. |

| QA ID | Area | Release-blocking assertion |
|---|---|---|
| QA18-055 | Security | Queue payloads reject unknown schema major version/job type/unknown fields before target access. |
| QA18-056 | Security | Workers never have implicit global scope because a site key is absent. |
| QA18-057 | Cross-doc | DOC-16 ranking/sitemap/redirect algorithms are consumed unchanged; worker mechanics do not re-specify them. |
| QA18-058 | Cross-doc | DOC-17 fixed 16-key variant catalog and private-original/native_webp rules are consumed unchanged. |
| QA18-059 | Cross-doc | DOC-15 operation count/method/path/status/body contracts remain unchanged. |
| QA18-060 | Cross-doc | No jobs/cache/outbox database table/column is added to DOC-08. |
| QA18-061 | Freeze gate | Cross-document audit reports zero unresolved blocking contradictions for DOC-18 scope. |
| QA18-062 | Freeze gate | All canonical job/queue/retry/Redis registries are internally unique with no duplicate IDs. |
| QA18-063 | Freeze gate | A same-slug/same-object-ID-across-sites test cannot collide in Redis, queue dedupe, search, sitemap or cache invalidation. |
| QA18-064 | Freeze gate | A worker restart/redelivery test converges to the same canonical/derived outcome without duplicate business mutation. |
| QA18-065 | Search dedupe | Two SEARCH-SYNC-POST triggers for the same site/post but different source_fingerprint values both remain enqueue-eligible; a prior 24-hour dedupe marker cannot suppress the newer source change. |
| QA18-066 | Search dedupe | SEARCH-REFRESH-ENTITY uses the exact Section 11.1 entity source_fingerprint; no undefined `source-update` placeholder remains in any registry or payload rule. |
| QA18-067 | Enqueue recovery | A failed enqueue with no durable broker acceptance compare-deletes its matching CLAIMED record; a producer crash/ambiguous result leaves only a 60-second provisional claim and reconciliation can recover afterward. |
| QA18-068 | Enqueue recovery | Only positive durable broker acceptance or actual worker receipt can establish ENQUEUED; a live CLAIMED marker is never permanent proof that equivalent work exists. |
| QA18-069 | Context | Global scheduler/maintenance coordination uses the frozen DOC-04 CTX-GLOBAL name only; all site-owned reads/writes execute under CTX-WORKER-SITE. |

# 36. Dependencies, Downstream Handoff and Freeze-Readiness

## 36.1 Downstream authorities

| Later PDF | May define | May not change without reopening DOC-18 |
|---|---|---|
| PDF-19 Security | Queue/Redis credentials, secret storage, network policy, security event retention, incident response. | Tenant/job identity, no-secret payload rule, retry/DLQ site safety. |
| PDF-20 Performance/Scale/DR/Observability | Worker counts, queue throughput, alert thresholds, Redis HA/backup choices, RPO/RTO, load tests. | At-least-once/idempotency/concurrency/retry correctness, cache isolation. |
| PDF-21 QA | Executable test harness and broader release suite. | Acceptance assertions defined here. |
| PDF-23 Engineering Architecture | Queue library/broker, Redis client, search engine, artifact store, worker repository/folder/runtime packages. | Logical queue/job/key contracts and authority boundaries. |
| PDF-24 Deployment | Process/container topology, environment wiring, rollout/rollback and migration/deployment sequencing. | No cross-tenant shared mutable scope or dropped queue durability requirements. |

## 36.2 Freeze-readiness checklist

- FZ18-01 Six logical queues and 11 canonical job types accepted.
- FZ18-02 CTX-WORKER-SITE plus frozen CTX-GLOBAL coordinator model accepted; no new trusted context name.
- FZ18-03 Common job envelope, exact search source_fingerprints and CLAIMED/ENQUEUED/TERMINAL dedupe state machine accepted.
- FZ18-04 Schedule generation tuple and 60-second reconcile accepted.
- FZ18-05 Scheduled publish stale/no-op and publish validation behavior accepted.
- FZ18-06 Media generation fingerprint and fixed-variant worker behavior accepted.
- FZ18-07 Incremental search + full site reindex serialization accepted.
- FZ18-08 Sitemap desired_seq/built_seq coalescing and atomic publish accepted.
- FZ18-09 Cache targeted vs SITE_PUBLIC_ALL invalidation model accepted.
- FZ18-10 Redis `bp:v1:s:{site_id}:` namespace/key registry accepted.
- FZ18-11 Redis TTL matrix and no Admin/Preview shared response caching accepted.
- FZ18-12 Retry/backoff/timeouts for seven profiles accepted.
- FZ18-13 Lease/token/fence concurrency rules accepted.
- FZ18-14 DLQ/replay tenant-safe rules accepted.
- FZ18-15 Maintenance reconciliation is non-destructive and derived-state-only.
- FZ18-16 Logging/metrics field semantics accepted.
- FZ18-17 No database/API/permission/page/feature expansion introduced.
- FZ18-18 Cross-document audit has zero unresolved blockers.
- FZ18-19 69 QA18 acceptance assertions accepted as release-blocking baseline.
- FZ18-20 Later PDFs are constrained to implementation/security/performance/deployment detail without redefining DOC-18 mechanics.

**REVIEW STATUS**

DOC-18 v1.0-RC2 is a controlled correction of RC1. The RC1 search-dedupe, enqueue-recovery and context-naming issues are resolved, and the refreshed cross-document audit has no known unresolved blocker for DOC-18 responsibilities. Formal freeze still requires explicit project approval.

# Appendix A. Canonical Job Registry

| Job ID | Queue | Dedupe identity basis | Retry profile |
|---|---|---|---|
| JOB18-PUBLISH-DUE | q18.publish | site+post+version+due | R18-PUBLISH |
| JOB18-MEDIA-VARIANTS | q18.media | site+media+generation+variant-set | R18-MEDIA |
| JOB18-SEARCH-SYNC-POST | q18.search | site+post+search-source-fingerprint | R18-SEARCH |
| JOB18-SEARCH-REFRESH-ENTITY | q18.search | site+entity-type+entity-id+entity-source-fingerprint | R18-SEARCH |
| JOB18-SEARCH-REINDEX-SITE | q18.search | site+accepted-command/rebuild generation | R18-REINDEX |
| JOB18-SITEMAP-BUILD-SITE | q18.sitemap | site+desired_seq | R18-SITEMAP |
| JOB18-CACHE-INVALIDATE | q18.cache | site+mode+dependency-set hash | R18-CACHE |
| JOB18-SCHEDULE-RECONCILE-SITE | q18.maintenance | site+minute bucket | R18-MAINT |
| JOB18-MEDIA-RECONCILE-SITE | q18.maintenance | site+15min bucket | R18-MAINT |
| JOB18-SEARCH-RECONCILE-SITE | q18.maintenance | site+15min bucket | R18-MAINT |
| JOB18-SITEMAP-RECONCILE-SITE | q18.maintenance | site+15min bucket | R18-MAINT |

# Appendix B. Queue / Job / Retry Matrix

| Queue | Jobs | Priority | Concurrency boundary | DLQ |
|---|---|---|---|---|
| q18.publish | PUBLISH-DUE | P1 | site+post | dlq18.publish |
| q18.media | MEDIA-VARIANTS | P2/P3 | site+media+generation | dlq18.media |
| q18.search | SYNC-POST, REFRESH-ENTITY, REINDEX-SITE | P2/P4 | site+post; site-wide reindex | dlq18.search |
| q18.sitemap | SITEMAP-BUILD-SITE | P3 | site | dlq18.sitemap |
| q18.cache | CACHE-INVALIDATE | P0/P2 | site+dependency/mode | dlq18.cache |
| q18.maintenance | 4 reconcile jobs | P4 | site+family | dlq18.maintenance |

# Appendix C. Redis Key Registry

| Key ID | Pattern | TTL |
|---|---|---|
| RED18-EPOCH | bp:v1:s:{site_id}:epoch:public | none |
| RED18-PAGE | bp:v1:s:{site_id}:cache:page:e{epoch}:{host_hash}:{route_hash} | 300s |
| RED18-DATA | bp:v1:s:{site_id}:cache:data:e{epoch}:{page_id}:{identity_hash}:{query_hash} | 300s |
| RED18-OBJ | bp:v1:s:{site_id}:cache:obj:{entity_type}:{entity_id}:v{version_token} | 600s |
| RED18-DEP | bp:v1:s:{site_id}:dep:{dep_type}:{dep_id} | 900s |
| RED18-DEDUPE | bp:v1:s:{site_id}:dedupe:{job_type}:{identity_hash} | job-specific |
| RED18-LOCK | bp:v1:s:{site_id}:lock:{purpose}:{identity_hash} | timeout+60s |
| RED18-SEARCH-STATUS | bp:v1:s:{site_id}:search:status | none |
| RED18-SMAP-DESIRED | bp:v1:s:{site_id}:sitemap:desired_seq | none |
| RED18-SMAP-BUILT | bp:v1:s:{site_id}:sitemap:built_seq | none |
| RED18-G-LOCK | bp:v1:g:lock:{purpose} | purpose lease |

# Appendix D. Trigger-to-Job Matrix

| Frozen upstream event | DOC-18 derived work |
|---|---|
| Schedule/Reschedule | PUBLISH-DUE |
| Due schedule reconcile | PUBLISH-DUE |
| Publish/Scheduled Publish/Update Published | CACHE-INVALIDATE + SEARCH-SYNC-POST + SITEMAP-BUILD |
| Archive/Trash/Permanent Delete | CACHE-INVALIDATE + SEARCH-SYNC-POST(delete by current truth) + SITEMAP-BUILD |
| Restore to Draft | Ensure cache/search/sitemap remain absent through current-truth sync |
| Category/Tag/Author public change | CACHE-INVALIDATE + SEARCH-REFRESH-ENTITY + SITEMAP-BUILD |
| Media upload/replace after native_webp | MEDIA-VARIANTS; replace also CACHE-INVALIDATE/CDN purge |
| Media metadata/variant availability change | CACHE-INVALIDATE |
| Media delete/public eligibility revoke | CACHE-INVALIDATE + media CDN purge |
| Search reindex API command | SEARCH-REINDEX-SITE |
| Site ENABLE | SITE_PUBLIC_ALL invalidation/epoch + SEARCH reconcile/reindex + SITEMAP build |
| Site DISABLE/ARCHIVE | SITE_PUBLIC_ALL + search ineligibility/reconcile + sitemap suppression/build |

| Frozen upstream event | DOC-18 derived work |
|---|---|
| Canonical domain change | SITE_PUBLIC_ALL + search destination refresh/reindex + sitemap build |

# Appendix E. Failure / Retry Quick Matrix

| Condition | Retry? | Final state |
|---|---|---|
| Expected stale version/generation/state | No | ACK STALE_NOOP |
| Transient dependency/network/lock timeout | Yes | Retry profile, then DLQ |
| Publish business validation fails at due time | No normal transient loop | DLQ BUSINESS_INVARIANT; Post unchanged |
| Cross-site/missing site context | No | Immediate DLQ SECURITY_INVARIANT |
| Duplicate side effect already committed | No harmful repeat | Idempotent success/stale-noop |
| Queue acceptance unavailable for 202 command | N/A before enqueue | DOC-15 503; no false AcceptedCommand |
| Derived state lost | Rebuild/reconcile | Canonical PostgreSQL unchanged |

# Appendix F. One-Page Developer Execution Reference

**BEFORE EXECUTING ANY SITE JOB**

Validate envelope -> resolve site -> verify lifecycle -> establish CTX-WORKER-SITE/RLS -> reload target -> verify site ownership/current state/version/generation -> acquire required lease/fence -> perform idempotent side effect -> commit/verify -> emit same-site downstream work -> ACK.

| Concern | Rule |
|---|---|
| Truth | PostgreSQL/content state is canonical; queue/Redis/search/sitemap/media derivatives are rebuildable. |
| Delivery | At-least-once; ACK after commit or stale-noop. |
| Tenant | Every site key/job/dedupe/lock/invalidation includes site_id. CTX-GLOBAL only enumerates sites; tenant work uses CTX-WORKER-SITE. Missing scope fails closed. |
| Dedupe | Exact source generation identity; Redis CLAIMED (60s) -> ENQUEUED -> TERMINAL. CLAIMED alone never proves queue acceptance. |
| Schedule | Delayed job + 60s site reconcile; stale generation tuple = no-op. |
| Search | Incremental current-truth projection; full reindex one site with shadow activation + catch-up. |

| Concern | Rule |
|---|---|
| Sitemap | desired_seq/built_seq, one build/site, atomic generation. |
| Cache | `bp:v1:s:{site_id}:`; public TTL 300s; targeted dependency invalidation; broad site epoch. |
| Media | Current generation fingerprint; only DOC-17 fixed variants; private original never public. |
| Retry | Seven fixed profiles; capped exponential + 0-20% deterministic jitter. |
| DLQ | One per queue; replay is not force; revalidate tenant/state. |